WORKED SOLUTION

STEP III 1994, Question 14

Probability and Statistics • Geometrical Probability

STEP III 1994, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q14

The key idea

An angle inscribed in a circle is half the angle the opposite arc subtends at the centre. So the three angles of the triangle are just π times the three arcs into which P, Q and R cut the circle, measured as fractions of the circumference. The question becomes one about how a circle is broken into three random pieces, and that is a question about areas inside a triangle.

Solution

Setting up

Measure positions on the circle as fractions of the circumference, so the circle has "length" 1. By rotational symmetry the position of P doesn't matter, so put P at 0. Then Q and R are at independent positions uniformly distributed on $[0,1)$.

Let $U < V$ be the positions of Q and R in increasing order. The pair (U,V) is uniform on the region $0 < u < v < 1$, which has area $\frac{1}{2}$, so its joint density is 2 there. (Ties have probability 0, and so do all the boundary cases below.) The three arcs are

$$X = U, \qquad Y = V - U, \qquad Z = 1 - V, \qquad X + Y + Z = 1.$$

The change of variables $(u,v) \mapsto (x,y) = (u, v-u)$ has Jacobian 1, so (X,Y) has density 2 on the triangle

$$T = \{(x,y) : x > 0,\ y > 0,\ x + y < 1\},$$

which has area $\frac{1}{2}$. So every probability below is *twice an area* inside T.

Ask yourself Why is each angle of the triangle π times an arc?

The angle of the triangle at R is an inscribed angle standing on the arc PQ that does not contain R. By the inscribed angle theorem it is half the angle that arc subtends at the centre. An arc that is a fraction x of the circumference subtends $2\pi x$ at the centre,

so the angle at R is πx. The same holds at each vertex, so the angles of PQR are πX, πY and πZ (and their sum is π, as it should be). Therefore

$$\text{an angle exceeds } k\pi \iff \text{the corresponding arc exceeds } k.$$

Let A_X, A_Y and A_Z be the events $X > k$, $Y > k$ and $Z > k$. We want $\mathrm{P}(A_X \cup A_Y \cup A_Z)$.

One arc exceeding k

For $0 \le k \le 1$:

- A_X is the part of T with $x > k$: a triangle with legs $1-k$, area $\frac{1}{2}(1-k)^2$.
- A_Y is the part with $y > k$: by the same calculation, area $\frac{1}{2}(1-k)^2$.
- A_Z is the part with $x + y < 1 - k$: the corner triangle with legs $1-k$, area $\frac{1}{2}(1-k)^2$.

Doubling, $\mathrm{P}(A_X) = \mathrm{P}(A_Y) = \mathrm{P}(A_Z) = (1-k)^2$.

The case $\frac{1}{2} \le k \le 1$

At most one arc can exceed k: if two did, their total would be more than $2k \ge 1$, but all three arcs add up to 1. So the events A_X, A_Y, A_Z are mutually exclusive, and

$$\mathrm{P}(\text{some angle exceeds } k\pi) = 3(1-k)^2,$$

as required.

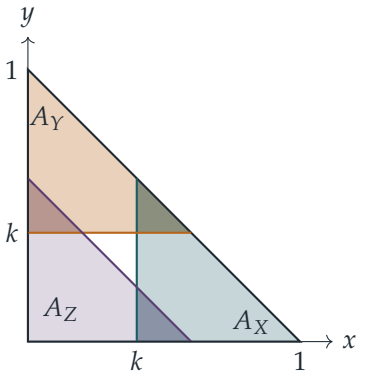
The case $\frac{1}{3} \le k \le \frac{1}{2}$

Now two arcs can both exceed k, so we use inclusion–exclusion.

Two arcs. $A_X \cap A_Y$ is the part of T with $x > k$, $y > k$, $x + y < 1$. Substituting $x = k + s$, $y = k + t$, this is $s, t > 0$, $s + t < 1 - 2k$, a triangle with legs $1 - 2k \ge 0$ and area $\frac{1}{2}(1-2k)^2$. So $\mathrm{P}(A_X \cap A_Y) = (1-2k)^2$. The other two pairs give the same: for example $A_X \cap A_Z$ is $x > k$, $y > 0$, $x + y < 1 - k$, again a triangle with legs $1 - 2k$.

Three arcs. All three exceeding k would need $X + Y + Z > 3k \ge 1$, which is impossible. So the triple intersection has probability 0.

The figure shows T for $k = 0.4$: the three corner triangles are A_X, A_Y, A_Z, and the darker overlaps are the pairwise intersections.



By inclusion–exclusion,

$$\begin{aligned}\mathrm{P}(A_X \cup A_Y \cup A_Z) &= 3(1-k)^2 - 3(1-2k)^2 + 0 \\ &= 3\big[(1-k)-(1-2k)\big]\big[(1-k)+(1-2k)\big] \\ &= 3k(2-3k).\end{aligned}$$

So for $\frac{1}{3} \le k \le \frac{1}{2}$ the probability that at least one angle exceeds $k\pi$ is

$$3k(2-3k) = 6k - 9k^2.$$

Checks

- At $k = \frac{1}{2}$ both formulae give $\frac{3}{4}$, so they join up continuously. This is the classic result that a random inscribed triangle is obtuse with probability $\frac{3}{4}$.
- At $k = \frac{1}{3}$ we get $3 \cdot \frac{1}{3} \cdot 1 = 1$. That is right: the largest angle is always at least $\frac{\pi}{3}$, since the three angles add up to π.
- At $k = 1$ we get 0, as no angle can exceed π.
- A simulation of 4×10^5 random triangles, computing the angles directly from the coordinates, gave 0.9602 at $k = 0.4$ (formula 0.96) and 0.4783 at $k = 0.6$ (formula 0.48).

Common slips

- Using the full arc PQ instead of the arc *not containing* the third vertex. Of the two arcs PQ, only one is the arc opposite R.
- Treating X, Y, Z as independent uniforms. They are constrained by $X + Y + Z = 1$, and the honest way to handle that is the area picture.
- Applying $3(1-k)^2$ below $k = \frac{1}{2}$. For $k < \frac{1}{2}$ it double-counts triangles with two large angles, and at $k = \frac{1}{3}$ it would give $\frac{4}{3} > 1$.

Beyond STEP

The arcs (X, Y, Z) are uniformly distributed on the simplex $x + y + z = 1$. This is the Dirichlet distribution, and it describes the "spacings" left when random points cut up an interval or a circle. With n points the same inclusion–exclusion gives $\mathrm{P}(\text{largest spacing} > k) = \sum_j (-1)^{j+1} \binom{n}{j} (1 - jk)_+^{n-1}$, a formula going back to Whitworth. Grimmett and Welsh, *Probability: An Introduction*, has a good treatment of geometrical probability and joint densities.

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